NEKHOROSHEV STABILITY FOR QUASI-CONVEX HAMILTONIANS: BLUEPRINT (TRACK N)

SCOTT ARMSTRONG AND ZOHAR KOMARGODSKI

Abstract. A proof plan, step by step, for the three frozen statements of track N of the KAM formalization: the exponential stability estimate for strictly quasi-convex nearly integrable Hamiltonians with a complex Hessian bound (N-HEAD-C, after Pöschel 1993, Theorem 1), the exact integrable dynamics for zero perturbation (N-ZERO) and the nonsingular isoenergetic derivative (N-QC). Every result is labelled and cites its pinned source. This version contains Part 0 only; the plan of the remaining parts is in `PLAN.md`.

Contents

1. Conventions, definitions and the frozen statements 1
1.1. Conventions 1
1.2. Track-K definitions used 2
1.3. Frozen track-N definitions 2
1.4. The frozen theorems 4

1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track N (owner approval D20, 27 October 2026) and the eleven frozen definitions of `KAM/Frozen/Nekhoroshev/Defs/` together with the five track-K definitions they use. The Lean declarations are authoritative; the statements below are transcriptions of them, with the Lean names recorded under each. N-HEAD with real bounds only is *not* frozen (source gap F-N1, D17/D18/D20); the frozen headline is N-HEAD-C. The additions of D22 (N-PARTIAL-CONVEX, N-AVERAGING) are not yet frozen and are not part of this part. Results of track K's blueprint are cited as [K:`bp:p1:...`].

1.1. Conventions.

- $n \in \mathbb{N}$ is the number of degrees of freedom; N-HEAD-C assumes $n \geq 3$ (D17), N-ZERO and N-QC are stated for every n. A phase point is $z = (\theta, I)$ with angles θ and actions I, in this order.
- Ambient spaces are those of the Lean text: $\mathbb{R}^n = (\texttt{Fin}\ n \to \mathbb{R})$, $\mathbb{C}^n = (\texttt{Fin}\ n \to \mathbb{C})$, with the max norms $|v| = \max_i |v_i|$ on both. The embedding $\mathbb{R}^n \subset \mathbb{C}^n$ is componentwise. The norm $\|\cdot\|$ in every frozen statement is the max norm, with two exceptions that are Euclidean on purpose: orthogonality in quasi-convexity uses the dot product $\sum_i \omega_i v_i$, and the phase space of a solution is $\mathbb{R}^n \times \mathbb{R}^n$.
- Operator norms of linear and multilinear maps are the induced norms of these max norms. In particular $\|D^2 h(z)\|$ for complex h is the norm of $D^2 h(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ (Lean `fderiv` applied twice), i.e. $\sup_{|u|,|v|\leq 1} |D^2 h(z)[u, v]|$.
- Angles are *not* reduced modulo $\mathbb{Z}^n$ in the statements: $\theta \in \mathbb{R}^n$ or $\theta \in T_s \subset \mathbb{C}^n$, and 1-periodicity is a hypothesis on f (Definition 1.12). So the torus is $\mathbb{R}^n/\mathbb{Z}^n$ and the Fourier characters are $e^{2\pi i k \cdot \theta}$ as in track K ([K:`bp:p1:decay`], [K:`bp:p1:summation`]). Pöschel's

torus is $\mathbb{R}^n/2\pi\mathbb{Z}^n$ and his norms are Euclidean; the conversion is a separate step of the proof (Part 1 of `PLAN.md`).
- The complex strip is $T_s = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta_i| < s\ \forall i\}$ (Lean `KAM.complexStrip`), as in track K.

1.2. **Track-K definitions used.**

Definition 1.1 (Complex action neighbourhood)**.** For $B \subset \mathbb{R}^n$ and $r \in \mathbb{R}$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ |I - J| < r\}$ (max norm, J viewed in $\mathbb{C}^n$). Thus $B_r = \emptyset$ for $r \le 0$, and for B open B_r is open.
Frozen Lean declaration: `KAM.actionNbhd`
Source: BM11 p. 4 (D_s); track K, [K:`bp:p0:domains`].

Definition 1.2 (Frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $\omega_h(I) = \nabla h(I) \in \mathbb{R}^n$ is the vector with $\omega_h(I)_i = D\big(J \mapsto \operatorname{Re} h(J)\big)(I)[e_i]$, the real partial derivatives of $\operatorname{Re} h$ on $\mathbb{R}^n$ at I. It is 0 if $\operatorname{Re} h|_{\mathbb{R}^n}$ is not differentiable at I (Lean totality, Remark 1.18).
Frozen Lean declaration: `KAM.frequencyMap`
Source: BM11 p. 1; Pos93 p. 190 ($\omega(I) = \partial_I h(I)$).

Definition 1.3 (Hamiltonian vector field)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and $z = (\theta, I)$, $X_H(z) = \big((\partial_{I_i} H(z))_i,\ (-\partial_{\theta_i} H(z))_i\big)$, where $\partial_{I_i} H(z) = DH(z)[(0, e_i)]$ and $\partial_{\theta_i} H(z) = DH(z)[(e_i, 0)]$ are the Fréchet derivatives in the Lean sense (0 if H is not differentiable at z).
Frozen Lean declaration: `KAM.hamiltonianVectorField`
Source: BM11 p. 1; track K, [K:`bp:p0:hamiltonian`].

Definition 1.4 (Real Hamiltonian)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and real $z = (\theta, I)$, $H_{h,f}(z) = \operatorname{Re}\big(h(I) + f(\theta, I)\big)$, with θ, I viewed in $\mathbb{C}^n$.
Frozen Lean declaration: `KAM.realHamiltonian`
Source: BM11 p. 5 ($H = h + f$).

Remark 1.5 (Strip and neighbourhood notation). `KAM.complexStrip` $n\, s = T_s$ as above; it is empty for $s \le 0$.
Frozen Lean declaration: `KAM.complexStrip`

1.3. **Frozen track-N definitions.**

Definition 1.6 (Action ball)**.** $B(0, R) = \{I \in \mathbb{R}^n : |I| < R\}$, the open ball of the max norm; written `actionBall` $n\, R$.
Frozen Lean declaration: `KAM.Nekhoroshev.actionBall`
Source: BM11 p. 4.

Definition 1.7 (Complex domain)**.** $D_s(R) = T_s \times B(0, R)_s \subset \mathbb{C}^n \times \mathbb{C}^n$; `complexDomain` $n\, R\, s$. It is the period-one lift of BM11's D_s, with a single width s for angles and actions. It is open.
Frozen Lean declaration: `KAM.Nekhoroshev.complexDomain`
Source: BM11 p. 4 (D_s); ours for the lift.

Definition 1.8 (Real action domain)**.** $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : I \in B(0, R)_s\}$; `realActionDomain` $n\, R\, s$.
Frozen Lean declaration: `KAM.Nekhoroshev.realActionDomain`
Source: BM11 p. 4 (B_s).

Remark 1.9 (Computing $\mathcal{B}_s(R)$). Since $|I - J|$ for real I, J is the real max norm, $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : \exists J,\ |J| < R,\ |I - J| < s\} = B(0, R + s)$ for $R, s > 0$. This identity is ours and is proved from the definitions (it is not stated in the source); it shows that $B(0, R) \subset \mathcal{B}_s(R)$ and that $\mathcal{B}_s(R)$ is open.

Definition 1.10 (Real part of the integrable part)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $h_{\mathbb{R}}(I) = \operatorname{Re} h(I)$; `realPart` h.
Frozen Lean declaration: `KAM.Nekhoroshev.realPart`
Source: BM11 p. 4 ($h : B_s \to \mathbb{R}$).

Definition 1.11 (Analytic integrable part, $h \in A_s(D)$)**.** `IsAnalyticIntegrable` $R\,s\,h$ holds if

(i) h is complex differentiable on $B(0,R)_s$;
(ii) h is continuous on $\overline{B(0,R)_s}$ (closure in $\mathbb{C}^n$);
(iii) there is $A \geq 0$ with $|h(I)| \leq A$ for all $I \in B(0,R)_s$;
(iv) $\operatorname{Im} h(I) = 0$ for all $I \in \mathcal{B}_s(R)$;
(v) $h_{\mathbb{R}}$ is of class C^3 on $\mathcal{B}_s(R)$ (real Fréchet sense; the set is open).

No numerical bound on h is fixed. Clause (v) is redundant by holomorphy ([K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:real-restriction`]) and records the regularity that guards every real derivative.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticIntegrable`
Source: BM11 p. 4 ($A_s(D)$, integrable case, with continuity on the closure).

Definition 1.12 (Analytic perturbation of size ε)**.** `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ holds if $\varepsilon \geq 0$ and

(i) f is complex differentiable on $D_s(R)$ and continuous on $\overline{D_s(R)}$;
(ii) $f(\theta + k, I) = f(\theta, I)$ for all $(\theta, I) \in D_s(R)$ and $k \in \mathbb{Z}^n$;
(iii) $\operatorname{Im} f(\theta, I) = 0$ for all $\theta \in \mathbb{R}^n$ and $I \in \mathcal{B}_s(R)$;
(iv) $|f(z)| \leq \varepsilon$ for all $z \in D_s(R)$ (complex absolute value).

The size is measured by $\leq \varepsilon$, not the source's strict bound; the source is applied with 2ε. For $\varepsilon = 0$ this forces $f = 0$ on $D_s(R)$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticPerturbation`
Source: BM11 pp. 4–5 ($f \in A_s(D)$, $|f|_s < \varepsilon$); deviation: $\leq \varepsilon$ (ruling Q2, D18).

Definition 1.13 (Derivative bounds $B(M)$)**.** For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$ and $M \in \mathbb{R}$, `HasDerivativeBounds` $U\,h\,M$ holds if $M > 0$, U is open, $h_{\mathbb{R}} \in C^3(U)$, and for all $I \in U$, all $k \in \{1,2,3\}$ and all $(i_1, \ldots, i_k) \in \{1, \ldots, n\}^k$,

$$\left|D^k h_{\mathbb{R}}(I)[e_{i_1}, \ldots, e_{i_k}]\right| \leq M.$$

These are all ordered coordinate partials of orders one to three; for C^3 functions they coincide with the multi-index partials $\partial^k h$ of the source (symmetry, [K:`bp:p1:ext-schwarz`]). The bound is on coordinate partials, not on operator norms.
Frozen Lean declaration: `KAM.Nekhoroshev.HasDerivativeBounds`
Source: BM11 p. 4, $B(M)$.

Definition 1.14 (Strict quasi-convexity $QC(m)$)**.** For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$, `IsStrictlyQuasiConvex` $U\,h\,m$ holds if $m > 0$, U is open, $h_{\mathbb{R}} \in C^2(U)$, and for every $I \in U$: $\omega_h(I) \neq 0$ and

$$\sum_i \omega_h(I)_i v_i = 0 \implies m|v|^2 \leq D^2 h_{\mathbb{R}}(I)[v,v] \qquad (v \in \mathbb{R}^n),$$

with $|v|$ the max norm on the left and the Euclidean dot product in the hypothesis.
Frozen Lean declaration: `KAM.Nekhoroshev.IsStrictlyQuasiConvex`
Source: BM11 p. 4, $QC(m)$ (the source's norm on the right is the max norm, orthogonality is Euclidean).

Definition 1.15 (Complex Hessian bound)**.** `HasComplexHessianBound` $R\,s\,h\,M_{\mathbb{C}}$ holds if $M_{\mathbb{C}} > 0$, h is C^2 in the complex sense on $B(0,R)_s$ and $\|D^2 h(z)\| \leq M_{\mathbb{C}}$ for all $z \in B(0,R)_s$, the norm being that of $D(Dh)(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ for the max norm. Every entry $|\partial_i \partial_j h|$ is then $\leq M_{\mathbb{C}}$, and the Euclidean operator norm of the Hessian matrix is $\leq n M_{\mathbb{C}}$. This is not the spectral radius of a complex symmetric matrix.
Frozen Lean declaration: `KAM.Nekhoroshev.HasComplexHessianBound`
Source: Pos93 p. 190, "The Hessian of the integrable hamiltonian h" (conversion by n recorded in `docs/tracks/N/sources.md`, anchor POS190-HESS).

Definition 1.16 (Solutions on a time set)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $U \subset \mathbb{R}^n$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$, `IsSolutionOn` $H\,U\,J\,z$ holds if for every $t \in J$: $z(t)_2 \in U$, H is differentiable at $z(t)$, and z is differentiable at t (two-sided, also at endpoints of J, since z is defined on $\mathbb{R}$) with $z'(t) = X_H(z(t))$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn`
Source: BM11 Theorem 2.1 p. 5 and the maximal-solution argument p. 13.

Definition 1.17 (The integrable orbit)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $\theta_0, I_0 \in \mathbb{R}^n$ and $t \in \mathbb{R}$: $\gamma_{\theta_0,I_0}(t) = (\theta_0 + t\,\omega_h(I_0),\, I_0)$.
Frozen Lean declaration: `KAM.Nekhoroshev.integrableOrbit`
Source: BM11 p. 1 ($H(\theta, I) = h(I)$).

Remark 1.18 (Lean totality). The frozen definitions and theorems use `fderiv` (which is 0 at a point of non-differentiability), `iteratedFDeriv`, the operator norm, `rpow` ($\varepsilon^{\pm 1/(2n)}$), `Real.exp` and the max norm; there is no matrix inverse, no Bochner integral and no division by a quantity that may vanish. The guards are:

(a) ω_h and $D^2 h_{\mathbb{R}}$ are used only on the open set $\mathcal{B}_s(R)$ where $h_{\mathbb{R}}$ is C^3 (Definitions 1.11(v), 1.13, 1.14), so the derivatives are the genuine ones.
(b) $D^2 h$ for complex h is used only on $B(0,R)_s$, open, where h is C^2 (Definition 1.15).
(c) X_H is used only inside `IsSolutionOn`, which requires differentiability of H at every visited point; in N-ZERO and N-HEAD-C this is a proof obligation: $H_{h,f}$ is real-differentiable on $\mathcal{B}_s(R) \times \mathbb{R}^n \supset B(0,R) \times \mathbb{R}^n$ because h, f are holomorphic on open sets containing the real points ([K:`bp:p1:hamiltonian-real`]).
(d) $\varepsilon^{-1/(2n)}$ and $\varepsilon^{1/(2n)}$ occur only with $\varepsilon > 0$ in N-HEAD-C, and $2n \neq 0$.
(e) In N-QC the map $\Psi(I', \lambda) = (h_{\mathbb{R}}(I'), \lambda\omega_h(I'))$ is differentiable at (I, λ) because $h_{\mathbb{R}} \in C^2$ on the open set $U \ni I$; `fderiv` is then the true derivative.

1.4. **The frozen theorems.**

Theorem 1.19 (N-HEAD-C: exponential stability, complex Hessian bound)**.** *Let $n \geq 3$ and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$. Then there are $C, T, c, \varepsilon_0 > 0$ with*

$$C\,\varepsilon_0^{1/(2n)} \leq R/4$$

such that the following holds for every $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and $\varepsilon \in \mathbb{R}$ with

(H1) `IsAnalyticIntegrable` $R\,s\,h$ *(Definition 1.11);*
(H2) `HasComplexHessianBound` $R\,s\,h\,M_{\mathbb{C}}$ *(Definition 1.15);*
(H3) $\omega_{\min} \leq |\omega_h(I)|$ *for every $I \in B(0,R)$ (max norm);*
(H4) `IsStrictlyQuasiConvex` $\mathcal{B}_s(R)\,h\,m$ *(Definition 1.14);*
(H5) `HasDerivativeBounds` $\mathcal{B}_s(R)\,h\,M$ *(Definition 1.13);*
(H6) `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ *(Definition 1.12);*
(H7) $0 < \varepsilon \leq \varepsilon_0$.

For every $\theta_0 \in \mathbb{R}^n$ and $I_0 \in B(0, R/2)$, with $\tau := T\exp\bigl(c\,\varepsilon^{-1/(2n)}\bigr)$ and $H = H_{h,f}$, there is $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ with

(a) $z(0) = (\theta_0, I_0)$;
(b) `IsSolutionOn` $H\,B(0,R)\,[-\tau, \tau]\,z$;
(c) $|z(t)_2 - I_0| \leq C\varepsilon^{1/(2n)}$ *for all t with $|t| \leq \tau$;*
(d) (uniqueness) for every $L \in [0, \tau]$ and every $y : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ with $y(0) = (\theta_0, I_0)$ and `IsSolutionOn` $H\,B(0,R)\,[-L, L]\,y$: $y(t) = z(t)$ *for all $t \in [-L, L]$.*

The order of choice is: n; $R, s, M, m, M_{\mathbb{C}}, \omega_{\min}$; then C, T, c, ε_0, depending only on these seven data; then h, f, ε, then θ_0, I_0; the solution z may depend on all of them. There is no dependence on A in

Definition 1.11(iii). In (H4)–(H5) the set is $\mathcal{B}_s(R) = B(0, R+s)$ (Remark 1.9), in (H3) the smaller ball $B(0, R)$.

Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_nekhoroshev_complexBound`

Source: Pos93 Theorem 1, pp. 190–191 (with the conversions and anchors of `docs/tracks/N/sources.md`*); exponent $1/(2n)$ as BM11 p. 5 with $\delta = (2n(n-1))^{-1}$; ruling D18 Q2, Q4; deviation D20: the extra inputs $M_{\mathbb{C}}$ and $\omega_{\min}$.*

Source gap: The printed theorem is for a 2π-periodic torus, Euclidean norms and exceptional initial frequencies $\|\omega(I_0)\| \le mr_0/8$; the conversion to the frozen statement (rescaling $H \mapsto 2\pi H(\varphi/2\pi, I)$, choice of r_0, s_0 from the seven data, exclusion of the exceptional branch by $\omega_{\min}$, existence and uniqueness of the solution) is given only in `sources.md` *and is proved in Part 10 of* `PLAN.md`*.*

Theorem 1.20 (N-ZERO: exact integrable dynamics)**.** *Let $n \in \mathbb{N}$, $R, s > 0$, h with* `IsAnalyticIntegrable` $R\, s\, h$ *and f with* `IsAnalyticPerturbation` $R\, s\, 0\, f$*. For every $\theta_0 \in \mathbb{R}^n$ and $I_0 \in B(0, R/2)$, with $\gamma = \gamma_{\theta_0, I_0}$ (Definition 1.17):*

$$\gamma(0) = (\theta_0, I_0), \qquad \texttt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, \mathbb{R}\, \gamma, \qquad \gamma(t)_2 = I_0 \quad (t \in \mathbb{R}).$$

Frozen Lean declaration: `KAM.Nekhoroshev.zero_perturbation`

Source: Ours (supplement to BM11 Theorem 2.1, p. 5, whose strict bound excludes $\varepsilon = 0$); from $H = h(I)$, BM11 p. 1.

Theorem 1.21 (N-QC: nonsingular isoenergetic derivative)**.** *Let $n \in \mathbb{N}$, $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$ with* `IsStrictlyQuasiConvex` $U\, h\, m$*. For $I \in U$ and $\lambda > 0$, the derivative at (I, λ) of*

$$\Psi : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n, \qquad \Psi(I', \lambda') = \big(h_{\mathbb{R}}(I'),\ \lambda' \omega_h(I')\big)$$

is a bijective linear map $\mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n$. Explicitly it is $D\Psi(I, \lambda)[v, u] = \big(\omega_h(I) \cdot v,\ u\, \omega_h(I) + \lambda D^2 h_{\mathbb{R}}(I) v\big)$. No uniform radius of invertibility is asserted.

Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_isoenergetic_derivative`

Source: BM11 Lemma 3.3 and its proof, pp. 9–10 (only the derivative assertion).